

# COM Port DLL specification (TANI DLL)

## TANI DLL Technical Reference

### 1. overview

This DLL is a library for easily and safely handling COM ports (serial communication) from Excel (VBA), Delphi, and C/C++. It supports 32-bit and 64-bit environments and is designed for communication with Arduino/ESP32.

### 2. constant definition

#### ■ return value

```
TANI_OK           = 0
TANI_OK_LICENSE   = 1
TANI_ERR          = -1
RESULT_BLOCKED    = -2
TANI_ERR_LICENSE  = -10
```

#### ■ Communication settings

```
TANI_PARITY_NONE = 0
TANI_PARITY_ODD  = 1
TANI_PARITY_EVEN = 2

TANI_STOP_1 = 1
TANI_STOP_2 = 2

TANI_FLOW_NONE    = 0
TANI_FLOW_RTSCTS  = 1
TANI_FLOW_XONXOFF = 2
```

#### ■ license control

```
LICENSE_CACHE_MIN = 60
COMM_FAIL_MAX      = 5
COMM_FAIL_WINDOW   = 5 / 1440.0
COMM_BLOCK_TIME    = 60 / 1440.0
```

### 3. Error code details

| value | Content            | handle          |
|-------|--------------------|-----------------|
| 0     | success            | none            |
| 1     | Successful license | First time only |

| value | Content               | handle               |
|-------|-----------------------|----------------------|
|       | authentication        |                      |
| -1    | General error         | Check COM settings   |
| -2    | temporary block       | wait time            |
| -10   | License not certified | Execute CheckLicense |

## 4. DLL function specification

### ■ TANI\_CheckLicense

#### Delphi

```
function TANI_CheckLicense(LicenseKey: PAnsiChar): Integer; stdcall;
```

#### C/C++

```
typedef int (__stdcall *TANI_CheckLicense_t)(const char* key);
```

#### argument

| name       | Variable types | explanation      |
|------------|----------------|------------------|
| LicenseKey | const char*    | subscription key |

#### return value

| value  | explanation               |
|--------|---------------------------|
| 1      | Authentication successful |
| others | failure                   |

#### VBA header

```
#If VBA7 Then
  #If Win64 Then
    Private Declare PtrSafe Function TANI_CheckLicense Lib
"TaniDLL64.dll" ( _
        ByVal LicenseKey As String) As Long
  #Else
    Private Declare Function TANI_CheckLicense Lib "TaniDLL32.dll" ( _
        ByVal LicenseKey As String) As Long
  #End If
#Else
  Private Declare Function TANI_CheckLicense Lib "TaniDLL32.dll" ( _
        ByVal LicenseKey As String) As Long
#End If
```

#### VBA Usage example

```
Dim ret As Long
ret = TANI_CheckLicense("XXXXXXXX")
If ret <> 1 Then
    MsgBox "license error", vbCritical
```

```
Exit Sub
End If
```

### Delphi Usage example

```
var ret: Integer;
begin
  ret := TANI_CheckLicense('XXXXXXXX');
end;
```

### C/C++ Usage example

```
int ret = TANI_CheckLicense("XXXXXXXX");
```

## ■ TANI\_OpenEx

### Delphi

```
function TANI_OpenEx(
  id: Integer;
  name: PAnsiChar;
  BAUD: Integer;
  databits: Integer;
  parity: Integer;
  stopbits: Integer;
  flow: Integer
): Integer; stdcall;
```

### C/C++

```
typedef int (__stdcall *TANI_OpenEx_t)(int, const char*, int, int, int, int, int);
```

### argument

| name     | Variable types | explanation    |
|----------|----------------|----------------|
| id       | Int            | Port ID (0-15) |
| name     | const char*    | COM port name  |
| BAUD     | Int            | baud rate      |
| databits | Int            | data bit       |
| parity   | Int            | parity         |
| stopbits | Int            | stop bit       |
| flow     | Int            | flow control   |

### return value

| value | Explanation |
|-------|-------------|
| 0     | success     |
| -1    | error       |

## VBA header

```
#If VBA7 Then
  #If Win64 Then
    Private Declare PtrSafe Function TANI_OpenEx Lib "TaniDLL64.dll" ( _
      ByVal id As Long, _
      ByVal name As String, _
      ByVal BAUD As Long, _
      ByVal databits As Long, _
      ByVal parity As Long, _
      ByVal stopbits As Long, _
      ByVal flow As Long) As Long
  #Else
    Private Declare Function TANI_OpenEx Lib "TaniDLL32.dll" ( _
      ByVal id As Long, _
      ByVal name As String, _
      ByVal BAUD As Long, _
      ByVal databits As Long, _
      ByVal parity As Long, _
      ByVal stopbits As Long, _
      ByVal flow As Long) As Long
  #End If
#Else
  Private Declare Function TANI_OpenEx Lib "TaniDLL32.dll" ( _
    ByVal id As Long, _
    ByVal name As String, _
    ByVal BAUD As Long, _
    ByVal databits As Long, _
    ByVal parity As Long, _
    ByVal stopbits As Long, _
    ByVal flow As Long) As Long
#End If
```

## VBA Usage example

```
Dim ret As Long
ret = TANI_OpenEx(0, "COM5", 115200, 8, 0, 1, 0)
If ret <> 0 Then
  MsgBox "open failure", vbCritical
End If
```

## Delphi Usage example

```
if TANI_OpenEx(0,'COM5',115200,8,0,1,0) <> 0 then Exit;
```

## C/C++ Usage example

```
if (TANI_OpenEx(0,"COM5",115200,8,0,1,0) != 0) return;
```

---

## ■ TANI\_Close

### Delphi

```
procedure TANI_Close(id: Integer); stdcall;
```

### C/C++

```
typedef void (__stdcall *TANI_Close_t)(int);
```

## argument

| name | Variable types | explanation    |
|------|----------------|----------------|
| id   | int            | Port ID (0-15) |

## return value

なし

## VBA header

```
#If VBA7 Then
  #If Win64 Then
    Private Declare PtrSafe Sub TANI_Close Lib "TaniDLL64.dll" ( _
      ByVal id As Long)
  #Else
    Private Declare Sub TANI_Close Lib "TaniDLL32.dll" ( _
      ByVal id As Long)
  #End If
#Else
  Private Declare Sub TANI_Close Lib "TaniDLL32.dll" ( _
    ByVal id As Long)
#End If
```

## Usage example

```
TANI_Close 0
```

```
TANI_Close(0); // Delphi / C/C++
```

## ■ TANI\_SendText

### Delphi

```
function TANI_SendText(id: Integer; txt: PAnsiChar; term: PAnsiChar):  
Integer; stdcall;
```

### C/C++

```
typedef int (__stdcall *TANI_SendText_t)(int, const char*, const char*);
```

## argument

| name | Variable types | explanation           |
|------|----------------|-----------------------|
| id   | int            | Port ID (0-15)        |
| txt  | const char*    | Send string           |
| term | const char*    | terminating character |

## return value

| value | explanation          |
|-------|----------------------|
| >0    | Number of bytes sent |
| <=0   | error                |

## VBA header

```
#If VBA7 Then
  #If Win64 Then
    Private Declare PtrSafe Function TANI_SendText Lib "TaniDLL64.dll"
  ( _
      ByVal id As Long, _
      ByVal txt As String, _
      ByVal TERM As String) As Long
  #Else
    Private Declare Function TANI_SendText Lib "TaniDLL32.dll" ( _
      ByVal id As Long, _
      ByVal txt As String, _
      ByVal TERM As String) As Long
  #End If
#Else
  Private Declare Function TANI_SendText Lib "TaniDLL32.dll" ( _
      ByVal id As Long, _
      ByVal txt As String, _
      ByVal TERM As String) As Long
#End If
```

## VBA Usage example

```
TANI_SendText 0, "HELLO", vbCr
```

## Delphi Usage example

```
TANI_SendText(0, 'HELLO', #13);
```

## C/C++ Usage example

```
TANI_SendText(0, "HELLO", "\r");
```

## ■ TANI\_SendBytes

### Delphi

```
function TANI_SendBytes(id: Integer; data: PByte; length: Integer): Integer;
stdcall;
```

### C/C++

```
typedef int (__stdcall *TANI_SendBytes_t)(int, unsigned char*, int);
```

## argument

| Name   | Variable types | explanation    |
|--------|----------------|----------------|
| Id     | int            | Port ID (0-15) |
| Data   | unsigned char* | <b>Data</b>    |
| Length | int            | Size           |

## return value

| Value | explanation          |
|-------|----------------------|
| >0    | Number of bytes sent |
| <=0   | error                |

## VBAheader

```
#If VBA7 Then
  #If Win64 Then
    Private Declare PtrSafe Function TANI_SendBytes Lib "TaniDLL64.dll"
    ( _
        ByVal id As Long, _
        ByRef data As Byte, _
        ByVal length As Long) As Long
  #Else
    Private Declare Function TANI_SendBytes Lib "TaniDLL32.dll" ( _
        ByVal id As Long, _
        ByRef data As Byte, _
        ByVal length As Long) As Long
  #End If
#Else
  Private Declare Function TANI_SendBytes Lib "TaniDLL32.dll" ( _
        ByVal id As Long, _
        ByRef data As Byte, _
        ByVal length As Long) As Long
#End If
```

## VBA Usage example

```
Dim buf(2) As Byte
buf(0) = 1
buf(1) = 2
buf(2) = 3
Dim ret As Long
ret = TANI_SendBytes(0, buf(0), 3)
If ret <= 0 Then
    MsgBox "Sending error", vbCritical
End If
```

## Delphi Usage example

```
var buf: array[0..2] of Byte = (1,2,3);
TANI_SendBytes(0,@buf[0],3);
```

## C/C++ Usage example

```
unsigned char buf[3]={1,2,3};
TANI_SendBytes(0,buf,3);
```

## ■ TANI\_RecvLine

### Delphi

```
function TANI_RecvLine(id: Integer; buf: PByte; bufSize: Integer; timeout:
Integer; term: Byte): Integer; stdcall;
```

### C/C++

```
typedef int (__stdcall *TANI_RecvLine_t)(int,unsigned char*,int,int,unsigned char);
```

## argument

| name    | Variable types | explanation    |
|---------|----------------|----------------|
| id      | int            | Port ID (0-15) |
| buf     | unsigned char* | buffer         |
| bufSize | int            | Size           |
| timeout | int            | ms             |
| term    | unsigned char  | termination    |

## return value

| value | explanation              |
|-------|--------------------------|
| >0    | Number of bytes received |
| 0     | Timeout                  |
| <0    | error                    |

## VBA header

```
#If VBA7 Then
    #If Win64 Then
        Private Declare PtrSafe Function TANI_RecvLine Lib "TaniDLL64.dll"
        ( _
            ByVal id As Long, _
            ByRef buf As Byte, _
            ByVal bufSize As Long, _
            ByVal timeout As Long, _
            ByVal TERM As Byte) As Long
    #Else
        Private Declare Function TANI_RecvLine Lib "TaniDLL32.dll" ( _
            ByVal id As Long, _
            ByRef buf As Byte, _
            ByVal bufSize As Long, _
            ByVal timeout As Long, _
            ByVal TERM As Byte) As Long
    #End If
#Else
    Private Declare Function TANI_RecvLine Lib "TaniDLL32.dll" ( _
        ByVal id As Long, _
        ByRef buf As Byte, _
        ByVal bufSize As Long, _
        ByVal timeout As Long, _
        ByVal TERM As Byte) As Long
#End If
```

## VBA Usage example

```
Dim buf(255) As Byte
Dim ret As Long
ret = TANI_RecvLine(0, buf(0), 256, 1000, 13)
If ret > 0 Then
    Dim s As String
    s = StrConv(buf, vbUnicode)
    MsgBox "Recv: " & Left(s, ret)
```



```

ElseIf ret = 0 Then
    MsgBox "timeout"
Else
    MsgBox " Reception error", vbCritical
End If

```

### Delphi Usage example

```

var buf: array[0..255] of Byte;
TANI_RecvLine(0,@buf[0],256,1000,13);

```

### C/C++ Usage example

```

unsigned char buf[256];
TANI_RecvLine(0,buf,256,1000,13);

```

## ■ TANI\_RecvBytes

### Delphi

```

function TANI_RecvBytes(id: Integer; buf: PByte; size: Integer; timeout:
Integer): Integer; stdcall;

```

### C/C++

```

typedef int ( __stdcall *TANI_RecvBytes_t)(int,unsigned char*,int,int);

```

### argument

| name    | Variable types | explanation    |
|---------|----------------|----------------|
| id      | int            | Port ID (0-15) |
| buf     | unsigned char* | buffer         |
| size    | int            | read size      |
| timeout | int            | ms             |

### return value

| value | explanation              |
|-------|--------------------------|
| >0    | Number of bytes received |
| 0     | Timeout                  |
| <0    | error                    |

### VBA header

```

#If VBA7 Then
    #If Win64 Then
        Private Declare PtrSafe Function TANI_RecvBytes Lib "TaniDLL64.dll"
        ( _
            ByVal id As Long, _
            ByRef buf As Byte, _
            ByVal size As Long, _

```

```

        ByVal timeout As Long) As Long
#Else
    Private Declare Function TANI_RecvBytes Lib "TaniDLL32.dll" ( _
        ByVal id As Long, _
        ByRef buf As Byte, _
        ByVal size As Long, _
        ByVal timeout As Long) As Long
#End If
#Else
    Private Declare Function TANI_RecvBytes Lib "TaniDLL32.dll" ( _
        ByVal id As Long, _
        ByRef buf As Byte, _
        ByVal size As Long, _
        ByVal timeout As Long) As Long
#End If

```

### VBA Usage example

```

Dim buf(255) As Byte
Dim ret As Long
ret = TANI_RecvBytes(0, buf(0), 10, 1000)
If ret > 0 Then
    MsgBox "Number of bytes received: " & ret
ElseIf ret = 0 Then
    MsgBox "timeout"
Else
    MsgBox "Reception error", vbCritical
End If

```

### Delphi Usage example

```

var buf: array[0..255] of Byte;
TANI_RecvBytes(0,@buf[0],10,1000);

```

### C/C++ Usage example

```

unsigned char buf[256];
TANI_RecvBytes(0,buf,10,1000);

```

## ■ TANI\_RecvBytesUntil

### Delphi

```

function TANI_RecvBytesUntil(id: Integer; buf: PByte; bufSize: Integer;
    termByte: Byte; inclTerm: Integer; timeout: Integer): Integer; stdcall;

```

### C/C++

```

typedef int (__stdcall *TANI_RecvBytesUntil_t)(int,unsigned
char*,int,unsigned char,int,int);

```

### argument

| name | Variable types | explanation    |
|------|----------------|----------------|
| id   | int            | Port ID (0-15) |
| buf  | unsigned char* | buffer         |

| name     | Variable types | explanation           |
|----------|----------------|-----------------------|
| bufSize  | int            | Size                  |
| termByte | unsigned char  | termination           |
| inclTerm | int            | including termination |
| timeout  | int            | ms                    |

## return value

| value | explanation              |
|-------|--------------------------|
| >0    | Number of bytes received |
| 0     | Timeout                  |
| <0    | error                    |

## VBA header

```
#If VBA7 Then
    #If Win64 Then
        Private Declare PtrSafe Function TANI_RecvBytesUntil Lib
"TaniDLL64.dll" ( _
            ByVal id As Long, _
            ByRef buf As Byte, _
            ByVal bufSize As Long, _
            ByVal termByte As Byte, _
            ByVal inclTerm As Long, _
            ByVal timeout As Long) As Long
    #Else
        Private Declare Function TANI_RecvBytesUntil Lib "TaniDLL32.dll" ( _
            ByVal id As Long, _
            ByRef buf As Byte, _
            ByVal bufSize As Long, _
            ByVal termByte As Byte, _
            ByVal inclTerm As Long, _
            ByVal timeout As Long) As Long
    #End If
#Else
    Private Declare Function TANI_RecvBytesUntil Lib "TaniDLL32.dll" ( _
        ByVal id As Long, _
        ByRef buf As Byte, _
        ByVal bufSize As Long, _
        ByVal termByte As Byte, _
        ByVal inclTerm As Long, _
        ByVal timeout As Long) As Long
#End If
```

## VBA Usage example

```
Dim buf(255) As Byte
Dim ret As Long
ret = TANI_RecvBytesUntil(0, buf(0), 256, &H23, 1, 1000)
If ret > 0 Then
    MsgBox "Number of bytes received: " & ret
ElseIf ret = 0 Then
    MsgBox "timeout"
Else
    MsgBox "Reception error", vbCritical
End If
```

### Delphi Usage example

```
var buf: array[0..255] of Byte;  
TANI_RecvBytesUntil(0,@buf[0],256,$23,1,1000);
```

### C/C++ Usage example

```
unsigned char buf[256];  
TANI_RecvBytesUntil(0,buf,256,0x23,1,1000);
```

---

## 5. Full communication sample

---

```
unsigned char buf[256];  
  
if (TANI_OpenEx(0,"COM5",115200,8,0,1,0) != 0) return;  
  
TANI_SendText(0,"HELLO","\r");  
  
TANI_RecvLine(0,buf,256,2000,13);  
  
TANI_Close(0);
```

---

```
Sub TestSerial()  
  
    Dim ret As Long  
    Dim buf(255) As Byte  
  
    ' COM open  
    ret = TANI_OpenEx(0, "COM5", 115200, 8, 0, 1, 0)  
    If ret <> 0 Then  
        MsgBox "open error", vbCritical  
        Exit Sub  
    End If  
  
    ' send  
    ret = TANI_SendText(0, "HELLO", vbCr)  
    If ret <= 0 Then  
        MsgBox "send error", vbCritical  
        GoTo EXIT_PROC  
    End If  
  
    ' Recv  
    ret = TANI_RecvLine(0, buf(0), 256, 2000, 13)  
    If ret > 0 Then  
        Dim s As String  
        s = StrConv(buf, vbUnicode)  
        MsgBox "Recv: " & Left(s, ret)  
    ElseIf ret = 0 Then  
        MsgBox "timeout"  
    Else  
        MsgBox "Recv error", vbCritical  
    End If  
  
EXIT_PROC:  
    TANI_Close 0  
  
End Sub
```

---

## 6. supplement

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- ・ ASCII language support
- ・ Port ID (0-15) management
- ・ Pointer-safe design
- ・ Offline support